Applications of AI - Fall 2026

# Is RAG Still Needed?

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# Two Competing Context Strategies

## Long Context

Dump entire documents straight into prompt memory.

## Vector RAG

Pre-filter knowledge using chunk embeddings.

# Why We Are Asking The Question

**1,000,000+**

Active Token Window

# The Whole-Book Dilemma

### ✕ Vector Search Fails

Retrieves existing words; completely blind to omissions and cross-file diffs.

### ✓ Full Attention Wins

Joint self-attention across both source documents detects missing clauses.

# The Rereading Tax

O(1)

**RAG Ingestion**

Embed once; query cheaply.

O(N)

**Raw Long Context**

Reprocess entire corpus on every prompt.

# Attention Dilution

- **Lost-in-the-Middle Phenomenon**

  Retrieval recall deteriorates sharply when critical facts sit midway through 500k+ tokens.

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**RAG removes the haystack before attention fires.**

# The Scale Ceiling

**1M Context Window**
Transient working memory (RAM)

~3 MB

**Enterprise Knowledge Lake**
Wikis, repos, ticketers, databases

Petabytes

# When To Choose Which

## Use Long Context

**Bounded Docs:** Single books or contracts.

**Global Synthesis:** Cross-document gap analysis.

**Batch Analysis:** Latency is non-critical.

## Use Vector RAG

**Unbounded Lake:** Petabyte corpus.

**Sub-Second SLA:** Fast user queries.

**Access Control:** Strict tenant permissions.

# Hierarchical AI Architecture

LAYER 01

## Vector Filter

Narrows petabytes to top 5 files.

LAYER 02

## Full Ingestion

Loads those files completely.

LAYER 03

## Attention Engine

Executes full-context reasoning.

# Thank you